

Design of IOT Architecture for Battery Charging Management System

The project submitted to the
SRM University – AP, Andhra Pradesh
for the partial fulfillment of the requirements to award the degree of

Master of Technology

In

EMBEDDED SYSTEMS & IOT

School of Engineering and Sciences

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[May 2025]

CERTIFICATE

I hereby certify that the work which is being presented in the M.Tech. Dissertation entitled “**Design of IOT Architecture for Battery Charging Management System**”, in partial fulfillment of the requirements for the award of the **Master of Technology in Embedded Systems & IoT, Department of Electronics and Communication Engineering** is an authentic record of my own work carried out during a period from 05-08-2024 to 31-04-2025 under the supervision of **Dr. Sujith kalluri Ph.D, Associate Professor**, Department of Electronics and Communication Engineering.

The matter presented in this thesis has not been submitted for the award of any other degree elsewhere.

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ACKNOWLEDGEMENT

First of all, I express my gratitude to the Family, who blessed me with the zeal and enthusiasm to complete this research work successfully. I am extremely thankful to my supervisor Dr. Sujith kalluri Ph.D, Associate Professor, Department of Electronics and Communication Engineering, SRM University – AP, for his/her motivation and tireless efforts to help me to get deep knowledge of the research area and supporting me throughout the life cycle of my M. Tech. dissertation work.

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ABSTRACT

This research proposes a novel IoT-based architecture for electric vehicle (EV) battery charging management, addressing the critical demand for scalable and sustainable charging infrastructure. The system integrates solar energy, energy storage systems (ESS), and grid power through a Point of Common Coupling (PCC), optimizing energy distribution via a five-layer IoT architecture: Sensor, Microcontroller & Data Acquisition, Communication, Cloud Server, and User Interface layers. Real-time monitoring, predictive analytics, and automated decision-making enhance operational efficiency, achieving 88.6% energy-saving efficiency (1,608 kW/day from renewables) and 88.5% cost-saving efficiency (₹19,296/day savings). The architecture supports battery swapping stations and DC fast charging, generating ₹26,986/day in revenue while stabilizing grid loads. This work bridges gaps in EV infrastructure standardization and economic viability, offering a scalable solution for smart cities and industrial applications.

Keywords: IoT Architecture, Renewable Energy Integration, Battery Swapping, Smart Grid, Predictive Analytics, Energy Storage Systems, Real-time Monitoring.

Abbreviations

EVs : Electric Vehicles

ESS : Energy Storage Systems

IoT : Internet of Things

PCC : Point of Common Coupling

MPPT : Maximum Power Point Tracking

SOC : State of Charge

CI : Charging Infrastructure

RES : Renewable Energy Sources

BSS : Battery Swapping Stations

GP : Grid Power

PA : Predictive Analytics

SCI : Smart City Initiatives

BESS : Battery Energy Storage System

RTDM : Real-Time Data Monitoring

OM : Operational Metrics

AWS : Amazon Web Services